7 November 2018 (Wednesday 9am)

Experiment 5

Harmonic Oscillator Part I: Spring Oscillator

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(Abstract: 177) / (Body: 2120)

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Abstract

Quantifying the Oscillation Frequencies of a Free and a Damped Oscillator

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When a spring-mass system falls under the influence of gravity and a restoring force, the system undergoes simple harmonic motion. Depending on initial conditions, the system oscillates at different frequencies as a result of external forces. The purpose of this experiment is to quantify the frequency of a spring-mass system undergoing free simple harmonic oscillation (undamped) and oscillatory motion with a damping force affecting the system (damped). In this experiment, the source of the damping effect will originate from the eddy currents produced by a magnet oscillating within an aluminum tube. Using a force sensor and excel, a displacement over time graph will be examined to calculate the oscillation frequencies of the undamped and damped oscillations. Mathematically, the formulas and their components are derived to predict the oscillation frequencies of the undamped and damped oscillations. Using both methods to verify the values, the resulting resonant frequencies are calculated: $f_0 = (0.6897 \pm 0.0238)Hz$ and $f_{damped} = (0.6872 \pm 0.0236)Hz$.

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Introduction

Simple harmonic oscillation occurs as a sinusoidal pattern induced by a force on a mass and a directly proportional restoring force. This motion is mathematically represented by partial differential equations solved during the 17th century when Newton and Leibniz developed calculus and Jean le Rond d'Alembert proposed the first equation representing a vibrating string. This experiment studies harmonic motion during undamped and damped oscillations. To accomplish this goal, the experiment will record the voltage outputs of a force sensor and the time when a spring-mass system is dropped from a height with a spring apply a restoring force for undamped oscillation. Then, the spring-mass system is dropped in the same manner but within an aluminum tube to induce a damping force as a result of interactions between magnets inside the mass and the aluminum tube, which further result in damping by eddy currents on the surface of the aluminum tube. Based on the recorded and measured data for each type of oscillating motion, the values will indicate varying frequencies for the system. In particular, the goal of the experiment is to the quantify the resonant frequencies of a spring-mass system undergoing both types of oscillation.

Methods

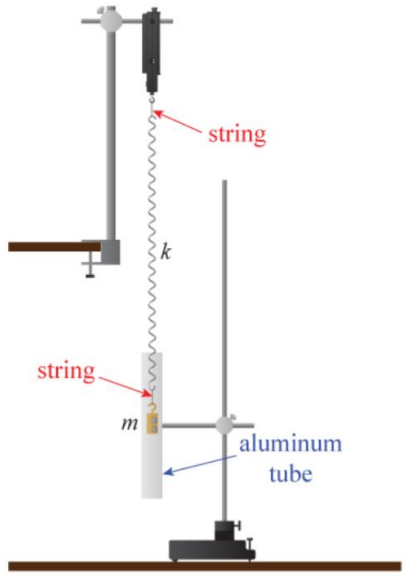

Figure 1.1. Experiment Setup for the undamped and damped oscillations. The undamped trial does not include the aluminum tube. A mass *m* oscillates after a experiencing a small external force while the force sensor records voltage measurements. The string serves to decouple the rotation of the spring.

The initial part of the experiment is the determination of the spring constant *k*. To determine the value of *k*, a spring was attached to the force sensor and the initial position was recorded. Four masses were placed on the spring and the displacement was recorded for each mass. By using the equation $F = mg$, the tension experienced by the mass-spring system was recorded for each mass. Then, a plot was created to compare the displacement of the system as the tension increased. The best-fit line yielded a linear equation which comprises of the spring constant *k* and the voltage offset due to the taring of the force sensor. With the values for the oscillating mass *m* and the calibration constant *k*, the predicted oscillation frequency is calculated using the definition of an undriven, undamped oscillation frequency:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

The first portion of the experiment quantifies the free oscillation frequency of an undamped oscillating mass *m*. With the setup indicated in figure 1.1 without the aluminum tube, the mass *m* was vertically displaced from its equilibrium position by a small magnitude to reduce uncertainty. With Capstone, time and voltage output results were recorded at a sample rate of 20 Hz for 30 seconds. In addition to predicting the free oscillation frequency, f_0 is calculated by plotting the data on a graph. Using two wave peaks, the time period of the oscillation is

calculated by finding the difference of the time positions of the peaks and dividing them by the difference of their wave peak index positions. By inverting the time period, the free oscillation frequency is calculated using empirical data.

The second portion of the experiment uses the same experiment as the first part except with an aluminum tube. The main idea behind the metal tube is to induce eddy currents on the surface of the aluminum tube as the mass *m* oscillates within the tube due to the magnetic flux produced by the change in the magnetic field, which results from the motion of the magnets within and along with the mass *m*. This portion of the experiment seeks to examine the damping effect due to magnetism and to quantify the damped oscillation frequency f_{damped}, the quality factor of the oscillation *Q*, and the damping time of the oscillation τ. By repeating the experiment procedures of the first portion of the experiment, the data for the damped oscillation frequency is plotted on a graph. Using the same method for calculating the oscillation frequency, the damped oscillation frequency f_{damped} is calculated. For the prediction of f_{damped}, the wave peak time positions and displacement positions are plotted on a graph. By finding the ratio of the successive amplitude measurements seen in the exponential decay graph for each period *T*, the damping time τ is calculated.

$$\frac{V(t+T)}{V(t)} = \frac{e^{-(t+T)/\tau}}{e^{-t/\tau}}$$

$$\frac{V(t+T)}{V(t)} = e^{\frac{-T}{t}}$$

$$\tau = -\frac{T}{ln\left[\frac{V(t+T)}{V(t)}\right]}$$

Using this value, the damping factor *b* and the quality factor *Q* is determined. This method predicts the value of the damped oscillation frequency f_{damped}.

Analysis

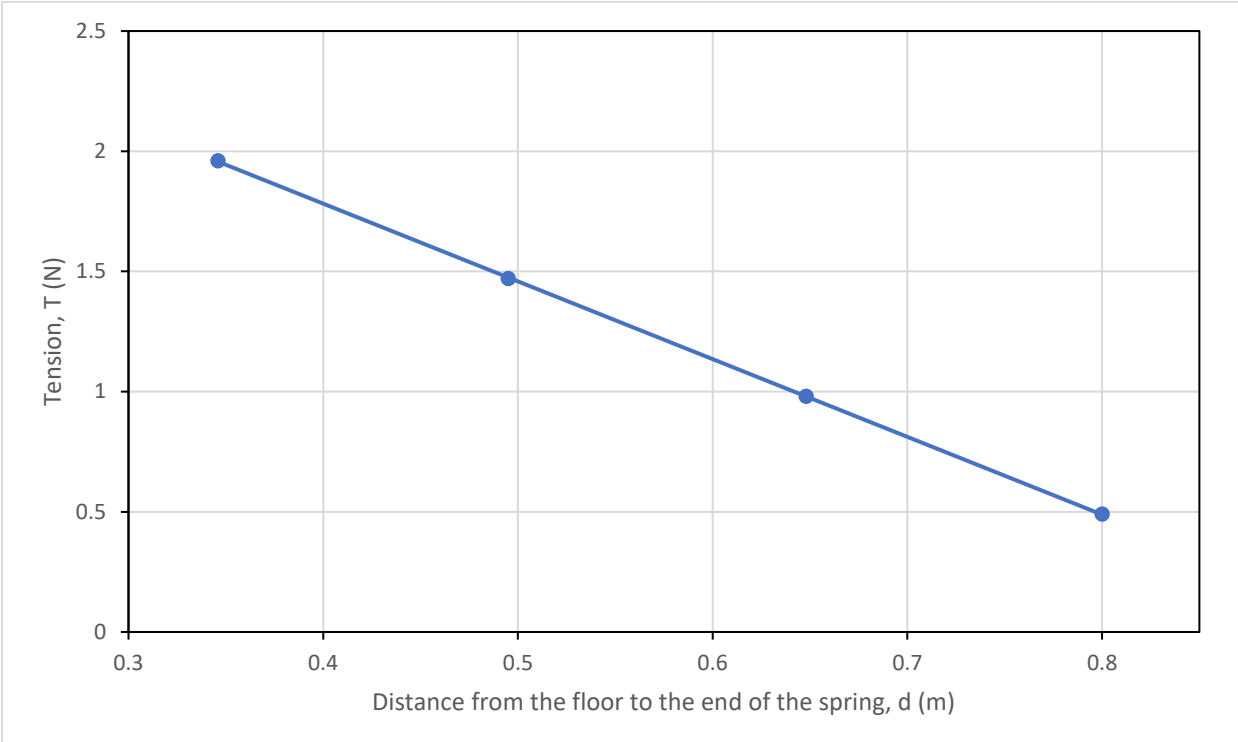

Figure 1.2. Spring constant determination. The tension was calculated for four masses (m = 50 g, 100 g, 150 g, 200 g) and the displacement of the spring for each mass was measured. These two values were plotted and the best-fit line yielded the linear equation $T = (-3.2342d + 3.0758)N$ and the spring constant $k = (3.2342 \pm 0.0126)N/m$.

With the spring constant and equation 5.1, the free oscillation frequency of an undriven, undamped oscillator (f_0) is quantified. Additionally, the uncertainty of f_0 is quantified using the uncertainty formulas ii.23 and ii.24.

$$\delta f_0 = |n| \left| \frac{1}{2\pi} \sqrt{\frac{k}{m}} \right| \sqrt{\left(\frac{\delta k}{|k_{best}|}\right)^2 + \left(\frac{\delta m}{|m_{best}|}\right)^2}$$

With the following values for the hanging mass $m = (0.17400 \pm 0.00005)g$, the spring constant $k = (3.2342 \pm 0.0126)N/m$, and the power factor $|n| = 1/2$, the predicted free oscillation frequency for the undriven, undamped oscillator (f_0) used in the spring constant determination and in the later parts of the experiment is determined.

$$f_{0,predicted} = (0.6862 \pm 0.0013)Hz$$

To verify the predicted free oscillation frequency of an undriven, undamped oscillator, the oscillatory motion of the hanging mass is plotted on a graph as voltage over time. As the hanging mass oscillates vertically, the force sensor records voltage values that correspond to the tension on the spring applied by the hanging mass m at time t. The free oscillation frequency of the oscillator is determined by inverting the time period of two wave peaks at T_i and T_j.

$$T = \frac{T_j - T_i}{n_{ij} - 1} \Rightarrow f_0 = \frac{1}{T} = \frac{n_{ij} - 1}{T_j - T_i}$$

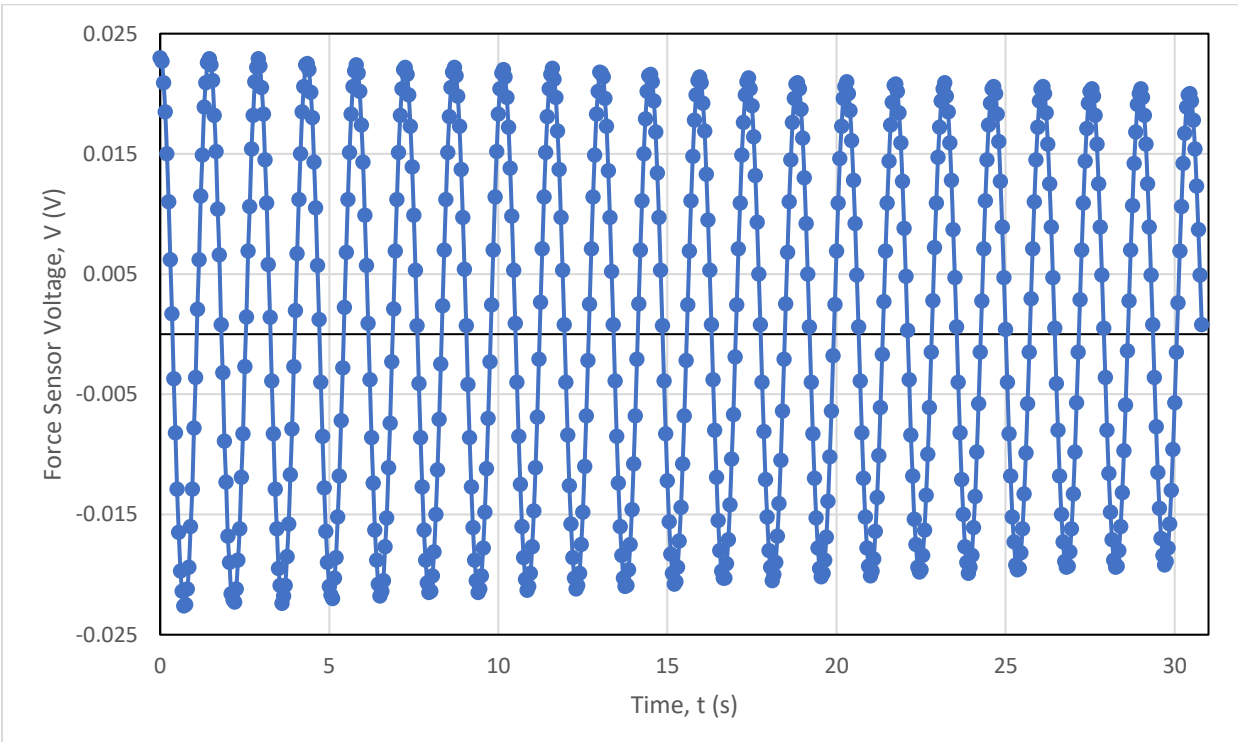

Figure 1.3. Undamped harmonic motion. The voltage output of the force sensor was plotted over time at a sample rate of 20 Hz as the hanging mass underwent simple harmonic motion.

With the definition of f_0 as the inverse of the time period, $f_{0,best}$ is calculated using the 1st and the 20th wave peaks and their respective time positions $t = (2.95, 30.5)s$. The uncertainty of the calculated free oscillation calculation is determined by the uncertainty formula ii.23 where $\delta T = 0.05$, Capstone's time difference for each data sample:

$$\delta f_0 = f_{0,best}\left(\frac{\delta T}{T_{best}}\right)$$

$$f_{0,calculated} = (0.6897 \pm 0.0238)Hz$$

The two predicted and calculated values for the free oscillation frequency of the undriven, undamped oscillating mass yield a difference of 0.0035 Hz, which lies in the margin of error for the second f_0 uncertainty. Therefore, the results indicate that the free oscillation frequency values agree with each other.

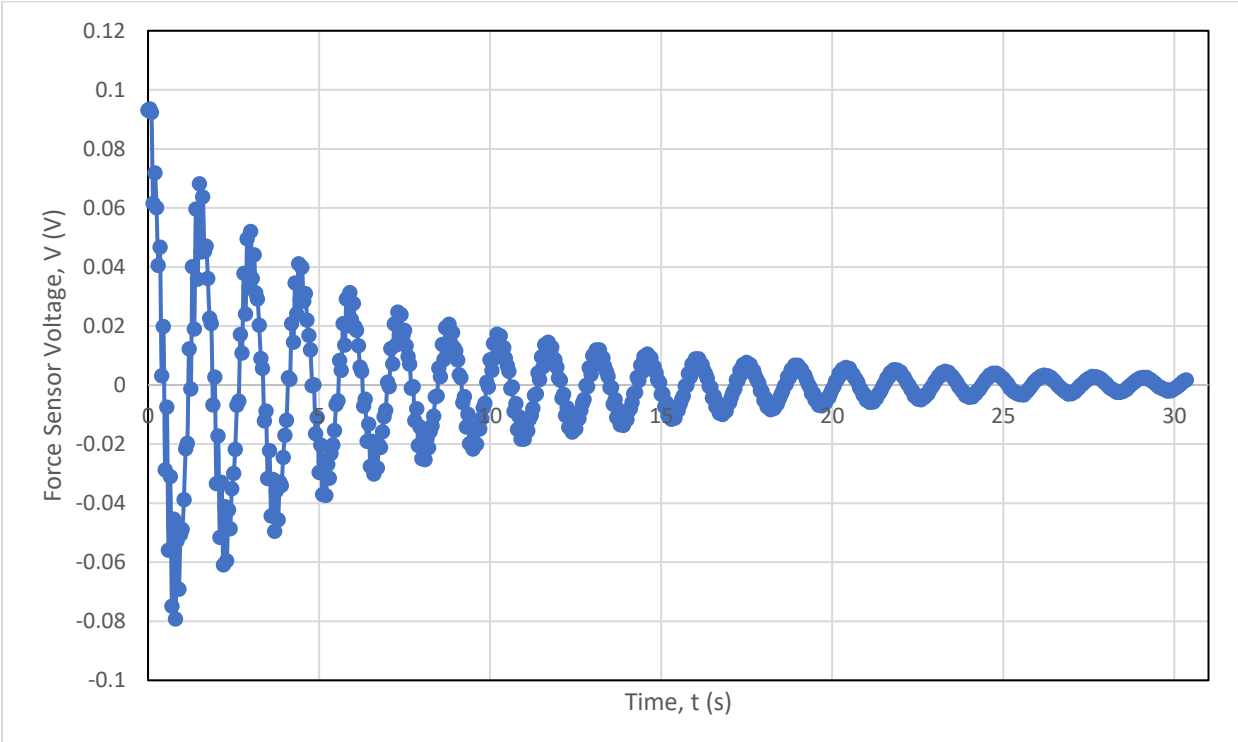

Figure 1.4. Damped harmonic motion. The voltage output of the force sensor was plotted over time at a sample rate of 20 Hz as the hanging mass underwent damped harmonic motion. The resulting graph was centered about the x axis by adding an offset of $V = (-0.0001t - 0.1005)V$ for each voltage data point V at time t, which was determined by the best-fit line of the graph.

By applying the same logic in the determination of the free oscillation frequency for an undriven, undamped oscillation, $f_{damped,best}$ and the uncertainty value is calculated by using the 1st and the 20th wave peaks and their time positions $t = (1.5, 29.15)s$. Mathematically, the value of f_{damped} is predicted using the formula indicated below where Q is the quality factor of the resonance. Additionally, the harmonic motion exhibits exponential decay, which is described as the damping time τ.

$$f_{damped,calculated} = (0.6872 \pm 0.0236)Hz$$

$$f_{damped,predicted} = f_0\sqrt{1 - \frac{1}{4Q^2}}$$

Derivation of the damping time τ

The damping time τ, defined as the amount of time for an undriven, damped oscillator to have its amplitude decrease by a factor of $1/e$, is calculated using the differential equation of the forces present in an undriven, damped oscillator and its complex solution. By substituting the complex solution and its derivations into the differential equation, the angular frequency of the oscillator is determined. By substituting the angular frequency into the complex solution, the damping time τ is determined.

Sum of forces:

$$\sum F = m\ddot{x} + kx + b\dot{x} = 0 \Rightarrow m\ddot{x} = -kx - b\dot{x}$$

Complex solution of an oscillator:

$$x(t) = Ae^{jwt}$$

Solution for ω:

$$-\omega^2 mAe^{jwt} = -kAe^{j\omega t} - j\omega Ae^{j\omega t}$$

$$m\omega^2 = k + j\omega b$$

$$\omega = \frac{jb}{2m} \pm \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

Imaginary and real solution of an undriven, damped oscillator:

$$x(t) = Ae^{j\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}t} \times e^{-bt/2m}$$

Solution for damping time τ:

$$\tau \equiv \frac{2m}{b}$$

Derivation of f_{damped}

The oscillation frequency of an undriven, damped oscillator (f_{damped}) is determined by dividing the angular frequency (ω) by 2π. By substituting the solution for ω as ω_{damped}, f_{damped} is determined.

$$f_{damped} = \frac{\omega_{damped}}{2\pi}$$

$$f_{damped} = \frac{1}{2\pi}\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$$f_{damped} = f_0\sqrt{1 - \frac{b^2}{4km}}$$

Derivation of the quality factor *Q*

The quality factor or Q-factor of the resonance of an undriven, damped oscillator is calculated using the definition of f_{damped} and *Q*. By equating f_{damped} and the definition of the *Q*, the quality factor *Q* is derived.

Definition of *Q*:

$$f_{damped} \equiv f_0\sqrt{1 - \frac{1}{4Q^2}}$$

Equation of f_{damped} and *Q*:

$$f_0\sqrt{1 - \frac{b^2}{4km}} = f_0\sqrt{1 - \frac{1}{4Q^2}}$$

Derivation of *Q*:

$$Q = \frac{\sqrt{km}}{b}$$

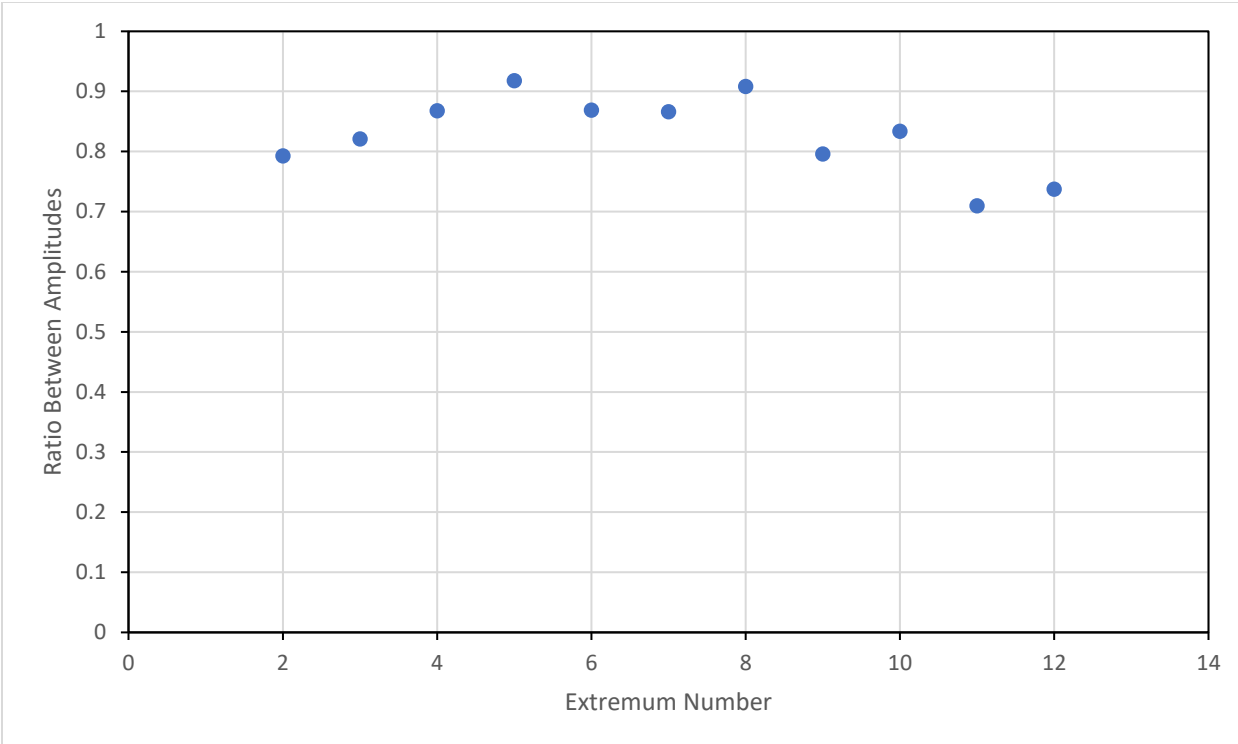

Figure 1.5. Analysis of the harmonic motion of a damped oscillator. The ratio for each consecutive wave peaks or maxima were plotted for each extremum until the 12th wave peak.

The results show a range from 0.709 to 0.918, which center around a mean of 0.815. The lack of a discernible slope show that the damping force is proportional to velocity since the wave function should decay at the same exponential rate regardless of amplitude. The mean damping time τ_{avg} is calculated using T and the ratio of successive amplitude measurements. The uncertainty was calculated using the procedure outlined in §ii.1.6.

$$\tau_{avg} = -\frac{T}{ln\left[\frac{V(t+T)}{V(t)}\right]} = (8.6097 \pm 1.2090)s$$

The quality factor Q is calculated by using the calculated τ to determine the damping factor b of the undriven, damped oscillation with its uncertainty calculated using the uncertainty formula ii.23. Then, Q is determined using the derivation of Q. The Q uncertainty is calculated using the uncertainty formulas ii.23 and ii.24 with $n = 1/2$.

$$\tau = \frac{2m}{b} \Rightarrow b = \frac{2m}{\tau} = (0.0404 \pm 0.0057)kg/s$$

Derivation of the uncertainty of *Q*:

Using the uncertainty formulas ii.23 and ii.24. First the square root of *k* and *m* are treated as one variable along with *b* to using the uncertainty formula for products and ratios. To compute for $\delta\sqrt{km}$, the uncertainty formula for quantities raised to a power where the uncertainty for the product of *k* and *m* are computed using the uncertainty formula for products and ratios.

$$\delta Q = Q_{best}\sqrt{\left(\frac{\delta\sqrt{km}}{\sqrt{km}}\right)^2 + \left(\frac{\delta b}{b}\right)^2}$$

$$\delta Q = Q_{best}\sqrt{\left(\frac{|n|\sqrt{\left(\frac{\delta k}{k}\right)^2 + \left(\frac{\delta m}{m}\right)^2}}{\sqrt{km}}\right)^2 + \left(\frac{\delta b}{b}\right)^2}$$

$$Q = \frac{\sqrt{km}}{b} = (18.5685 \pm 2.6203)$$

With the quality factor *Q* for the oscillation, the predicted damped oscillation frequency $f_{damped,predicted}$ is determined using the definition for *Q* and compared to the calculated damped oscillation frequency. The uncertainty of the value is also determined using the uncertainty formula for products and ratios ii.23.

$$f_{damped,predicted} = f_0\sqrt{1 - \frac{1}{4Q^2}}$$

$$f_{damped,predicted} = (0.6870 \pm 0.0998)Hz$$

	Predicted	Calculated
f_{damped}	0.6870 ± 0.0998 Hz	0.6872 ± 0.0236 Hz

Table 1.1. Predicted and Calculated damped oscillation frequencies. The results show that both values lie within the margins of error for either method of determination by a difference of 0.0002 Hz. Hence, the values for f_{damped} agree with each other regardless of the difference between the uncertainty values.

Conclusions

The experiment seeks to quantify and verify the resonant frequencies of a spring-mass system undergoing undamped and damped oscillations. With the mass value of the oscillating mass $m = (174.00 \pm 0.05)g$ and spring constant $k = (3.2342 \pm 0.0126)N/m$, the free oscillation frequency was predicted to be $f_{0,predicted} = (0.6862 \pm 0.0013)Hz$. By plotting the displacement of the undamped oscillations over 30 seconds as a wave function, the free oscillation frequency was calculated to be: $f_{0,calculated} = (0.6897 \pm 0.0238)Hz$. With a difference of 0.0035 Hz, the calculated resonant frequency for an undriven, undamped oscillator matches the predicted value within the margin of error. For the damped oscillation in which the damping source originated from eddy currents on the surface of the aluminum tube, the oscillation frequency was calculated using the same graph analysis method: $f_{damped,calculated} = (0.6872 \pm 0.0236)Hz$. By deriving and computing the damping time $\tau = (8.6097 \pm 1.2090)s$, then the damping force $b = (0.0404 \pm 0.0057)kg/s$, and then the quality factor $Q = (18.5685 \pm 2.6203)$, the oscillation frequency was calculated to be: $f_{damped,predicted} = (0.6870 \pm 0.0998)Hz$. Similar to the undamped oscillation calculations, with a difference of 0.0002 Hz, the calculated resonant frequency for an undriven, damped oscillator matches the predicted value within the margin of error.

Although the results match the expectations for the experiment, numerous improvements can be implemented into the experimentation process to reduce the uncertainty for any of the predicted and calculated values. For this experiment, uncertainty appeared most significantly in the determination of the damping time. The determination of the damping relied on the ratios of the successive amplitudes; hence, the uncertainty originated from the measurement of the data and the initial conditions of the oscillation that led to the data. For the method of measurement, the sampling rate should be more than 20 Hz to ensure that there are enough data points to properly produce a continuous wave function. In the graphs produced in the experiment, the wave function did not show enough data points to accurately show the local maxima for each half-cycle. This source of uncertainty would cause $f_{damped,predicted}$ to tend to shift to the lower half of the margin of error due to the direction relationship between the damping time and the quality factor. To prevent this uncertainty, the sampling rate should be increased.

Extra Credit

By using the Fast Fourier Transform tool in Capstone to determine the frequency response of the oscillating system, the width of the resonant frequency can be measured by determining the frequency values for which the amplitude drops to $1/\sqrt{2}$ of its initial value. The quality factor Q can be determined by finding the ratio of the resonant frequency f_0 and the width of the resonant frequency Δf: $Q = f_0/\Delta f$. For the measured FFT graph values, the amplitude was measured to be 0.01807 V and the decayed amplitude was calculated to be 0.01277 V. The amplitude corresponds to 0.68 Hz and the decayed amplitude corresponds to 0.63 Hz and 0.71 Hz.

$$Q = \frac{f_0}{\Delta f} = \frac{0.68}{|(0.71 - 0.63)|} = 8.5$$

Based on the results from the FFT graph and from the time-domain calculations, the quality factor values do not agree with each other. A notable source of error in the FFT, which may have resulted in the difference of 10.1, is the inaccurate graph. In the FFT graph produced in Capstone, the resonant frequency and the resonant width were both within the domain of the same data rectangle.

Bibliography

1. Campbell, W. C. et al. Physics 4AL: Mechanics Lab Manual (ver. June 27, 2018). (Univ. California Los Angeles, Los Angeles, California).